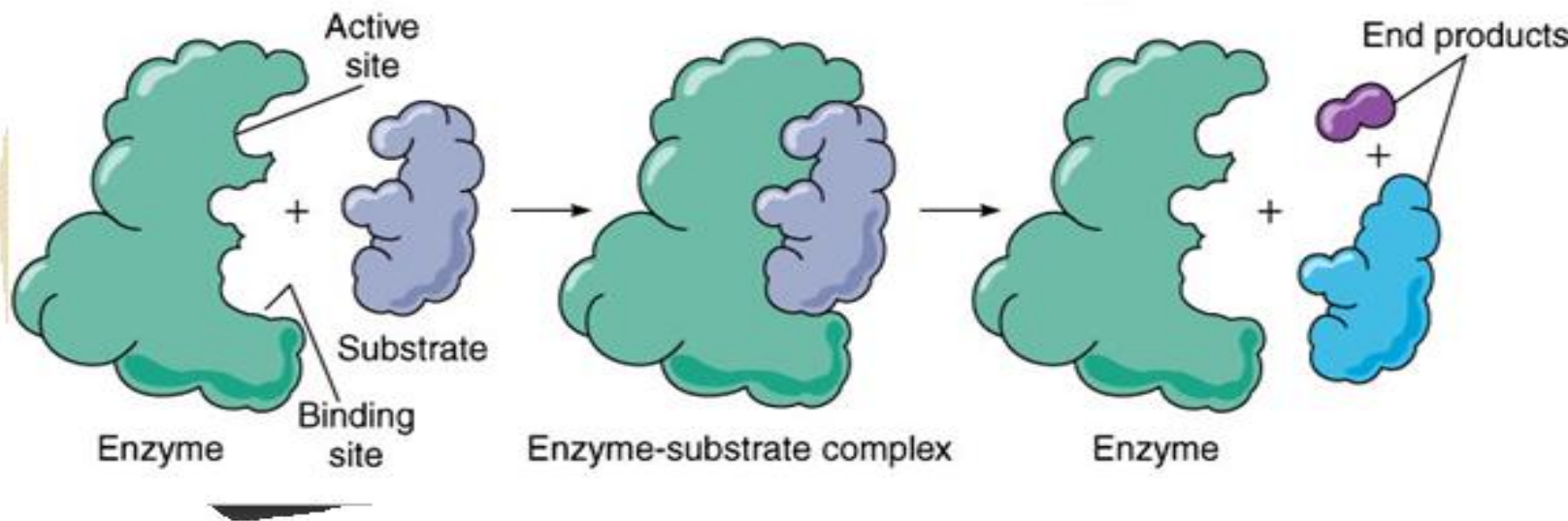




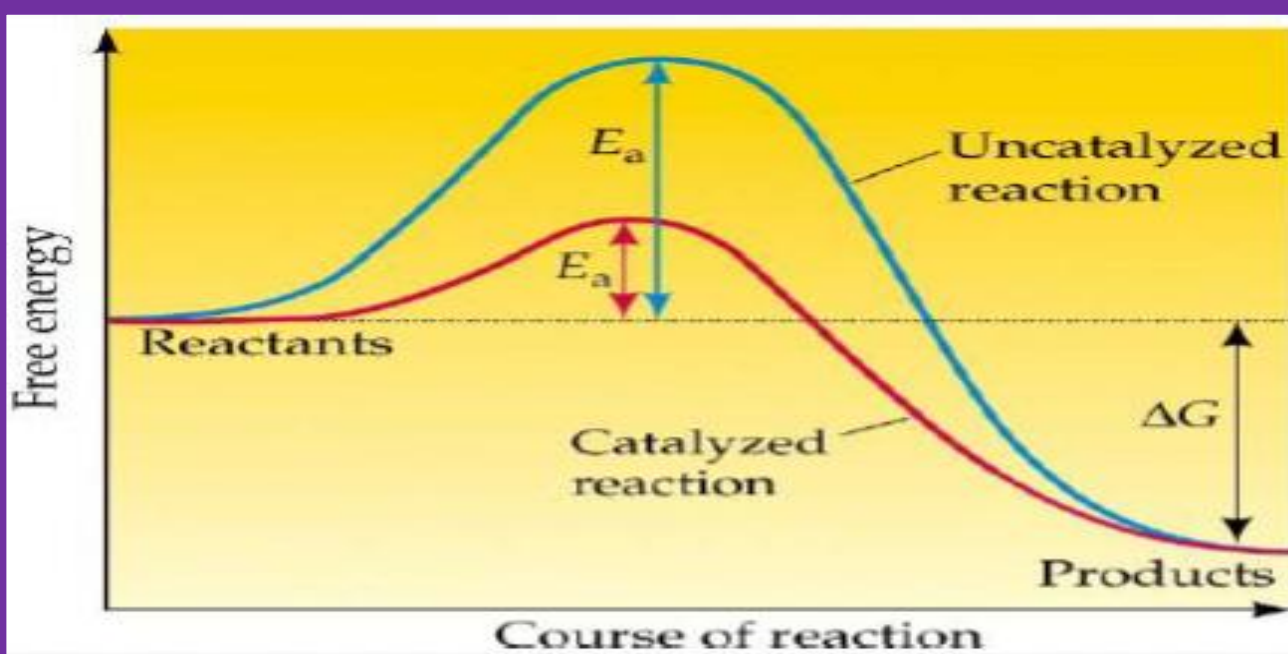
Enzyme

By: Alie M.

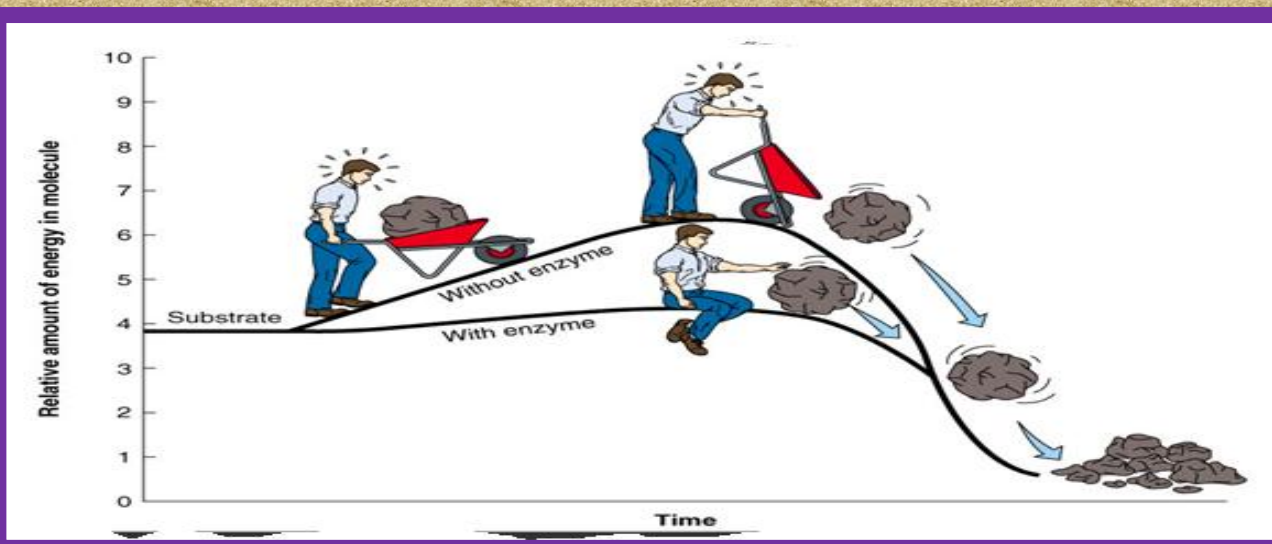
Outline

- ◆ Definition and basic structure of enzyme
- ◆ Classification of enzymes
- ◆ Factors affecting enzymatic activity
- ◆ Mechanism of enzymes
- ◆ Enzyme kinetics

■ **Enzymes** may be defined as biocatalysts synthesized by living cells. Almost all enzymes are proteins except catalytically active ribonucleic acids, the ribozymes. They can be described as any of several complex proteins that are produced by cells and act as catalysts in specific biochemical reactions. Biological catalysts are those catalysts which are found in living organisms and they speed up the metabolic reactions occurring in them. A substance that lowers activation energy of a reaction so the reaction occurs more quickly but, in the end, is not used up by the reaction is called a catalyst.



ACTIVATION ENERGY: The minimum amount of energy required to start a reaction is known as its activation energy. Activation Energy can be defined as the energy that must be added to cause molecules to react with one another.



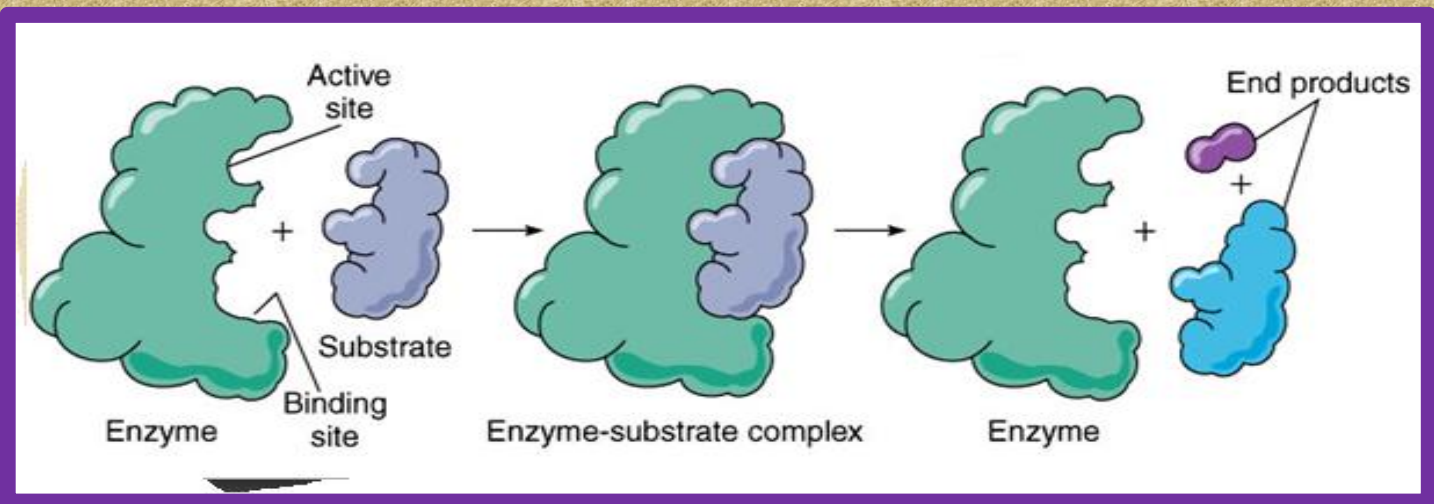
Enzymes act as catalysts.

- Found in cells throughout the body

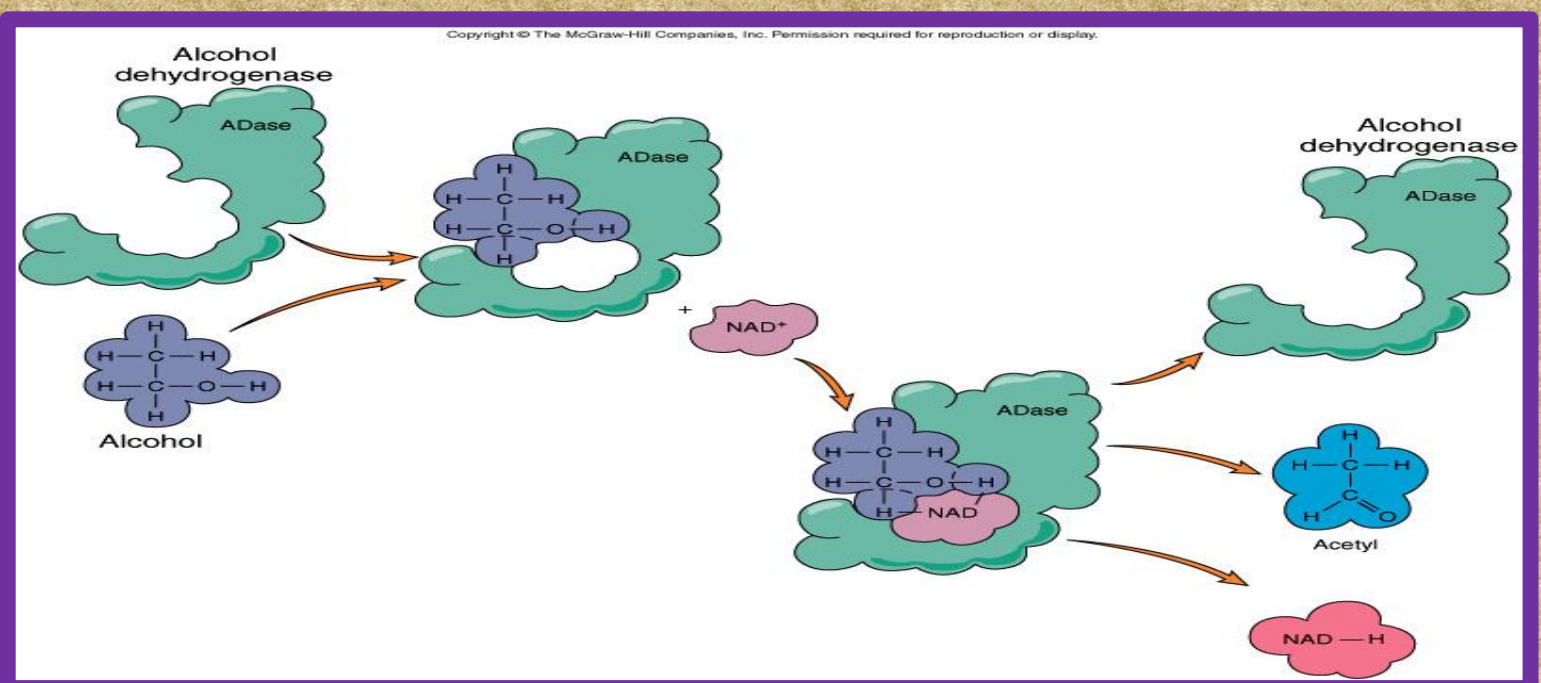
- Lowers activation energy

How Enzymes Work

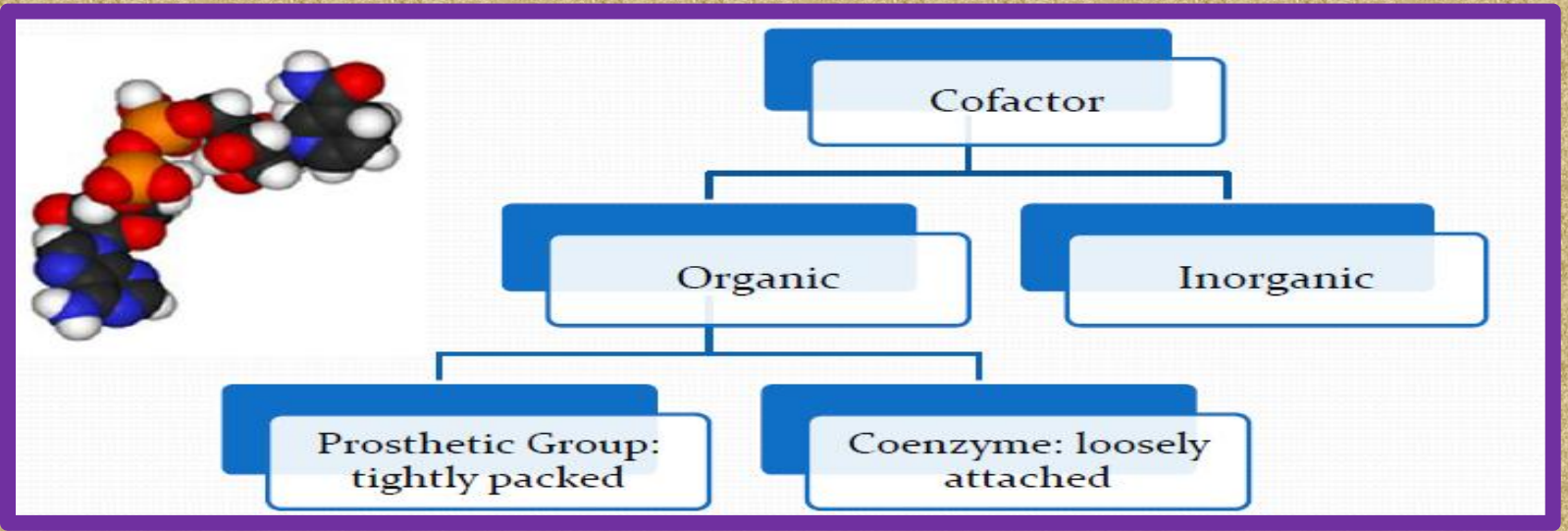
- ◆ Each enzyme has a specific size and 3-D shape
- ◆ Each enzyme is going to fit with a certain substrate (molecule enzyme connects to)



- ◆ When the enzyme and substrate are connected, it is known as **enzyme-substrate complex**. The **binding site** is where the enzyme physically attaches itself to the substrate. The **active site** is where the enzyme will cause a specific part of the substrate to change.
- ◆ **Cofactors/Coenzymes:** Some enzymes need an additional molecule to carry out the process. Co-factors are inorganic ions or organic molecules that serve an enzyme helpers. Co-enzymes are organic molecules that function as a cofactor.

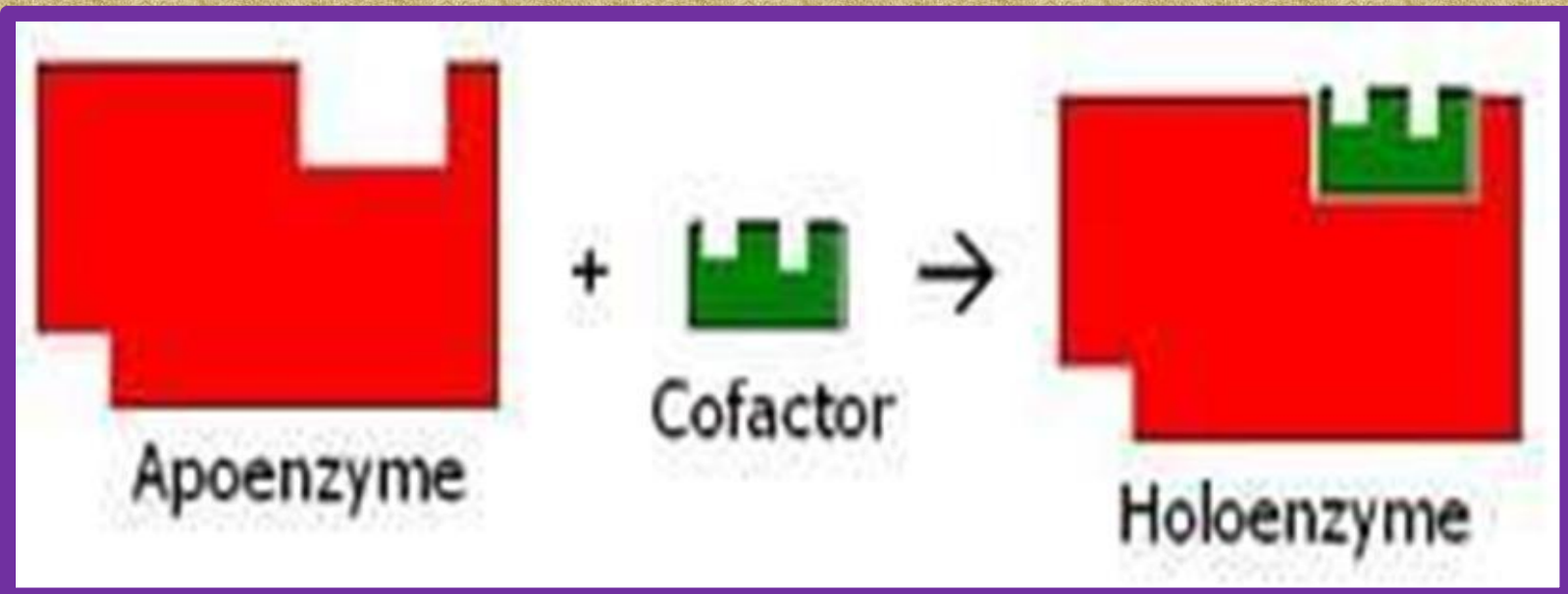


Classification of cofactors



Holoenzymes

- Some enzymes require molecules other than proteins for enzymatic activity. The term **holoenzyme** refers to the active enzyme with its nonprotein component. The term **apoenzyme** is inactive enzyme without its nonprotein part. If the nonprotein part is a metal ion such as Zn^{2+} or Fe^{2+} , it is called a cofactor. If it is a small organic molecule, it is termed a coenzyme.



Co-enzymes that only transiently associate with the enzyme are called **co-substrates**. It acts by donating or accepting hydrogen atoms (groups other hydrogen) or electrons (NAD^+ , $NADP$, FAD and FMN , ATP). If the coenzyme is permanently associated with the enzyme and returned to its original form, it is called a **prosthetic group** as FAD . Co-enzymes frequently are derived from vitamins.

Example, NAD^+ contains niacin, FAD contains riboflavin.

TABLE 6.3 Coenzymes of B-complex vitamins*

<i>Coenzyme (abbreviation)</i>	<i>Derived from vitamin</i>	<i>Atom or group transferred</i>
Thiamine pyrophosphate (TPP)	Thiamine	Aldehyde or keto
Flavin mononucleotide (FMN)	Riboflavin	Hydrogen and electron
Flavin adenine dinucleotide (FAD)	Riboflavin	"
Nicotinamide adenine dinucleotide(NAD ⁺)	Niacin	"
Nicotinamide adenine dinucleotide phosphate (NADP ⁺)	"	"
Lipoic acid	Lipoic acid	"
Pyridoxal phosphate (PLP)	Pyridoxine	Amino or keto
Coenzyme A (CoA)	Pantothenic acid	Acyl
Tetrahydrofolate (FH ₄)	Folic acid	One carbon (formyl, methenyl etc.)
Biocytin	Biotin	CO ₂
Methylcobalamin; Deoxyadenosyl cobalamin	Cobalamin	Methyl/isomerisation

Table 6.4 Coenzymes not related to B-complex vitamins

<i>Coenzyme</i>	<i>Abbreviation</i>	<i>Biochemical functions</i>
Adenosine triphosphate	ATP	Donates phosphate, adenosine and adenosine monophosphate (AMP) moieties.
Cytidine diphosphate	CDP	Required in phospholipid synthesis as carrier of choline and ethanolamine.
Uridine diphosphate	UDP	Carrier of monosaccharides (glucose, galactose), required for glycogen synthesis.
S - Adenosylmethionine (active methionine)	SAM	Donates methyl group in biosynthetic reactions.
Phosphoadenosine phosphosulfate (active sulfate)	PAPS	Donates sulfate for the synthesis of mucopolysaccharides.

Classification of Enzyme: Enzyme are classified on the basis of action it performs

- ◆ **Oxidoreductases**-oxidation-reduction reactions : Phosphate dehydrogenase
- ◆ **Transferases**-transfer of functional groups:Methyl transferases , Carboxyl transferases
- ◆ **Hydrolases**-hydrolysis reactions:Carboxylic ester hydrolases
- ◆ **Isomerases**-isomerization reactions:Epimerases
- ◆ **Lyases**-addition to double bonds:Carboxylyases, Aldehydelyases
- ◆ **Ligases**-formation of bonds with ATP cleavage:Amino acid-RNA ligases

Intracellular and extracellular enzymes

Intracellular enzymes :Are synthesized and retained in the cell for the use of cell itself.

They are found in the cytoplasm, nucleus, mitochondria and chloroplast.Example: Oxidoreductase catalyses biological oxidation, Enzymes involved in reduction in the mitochondria

Extracellular enzymes:Are synthesized in the cell but secreted from the cell to work externally.Example: Digestive enzyme produced by the pancreas, are not used by the cells in the pancreas but are transported to the duodenum.

Enzymes specificity:Enzymes are highly specific in nature, interacting with one or few substrates and catalyzing only one type of chemical reaction.Substrate specificity is due to complete fitting of active site and substrate.Example:Oxdo-reductase do not catalyze hydrolase reactions and hydrolase do not catalyze reaction involving oxidation and reduction.Enzymes show different degrees of specificity:

- ◆ **Bond specificity.**
- ◆ **Group specificity.**
- ◆ **Absolute specificity.**
- ◆ **Optical or stereo-specificity.**
- ◆ **Dual specificity.**

Bond specificity: In this type, enzyme acts on substrates that are similar in structure and contain the same type of bond.Example:•**Amylase** which acts on α -1-4 glycosidic bond in starch, dextrin and glycogen, shows bond specificity.**Lipase**: which acts only on ester bonds.

Absolute specificity: in this type, an enzyme act only on one substrate.Example;

Uricase; acts only on urinic acid.

Carbonic anhydrase : acts on carbonic acid.

Arginase : acts on arginine

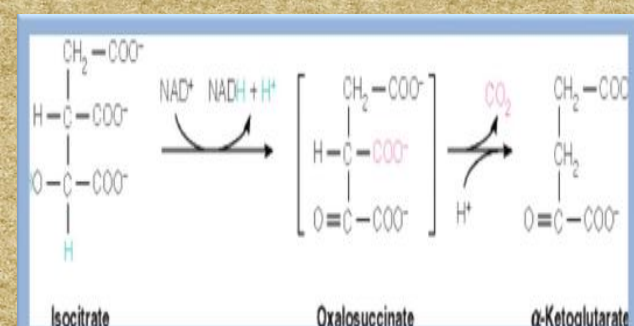
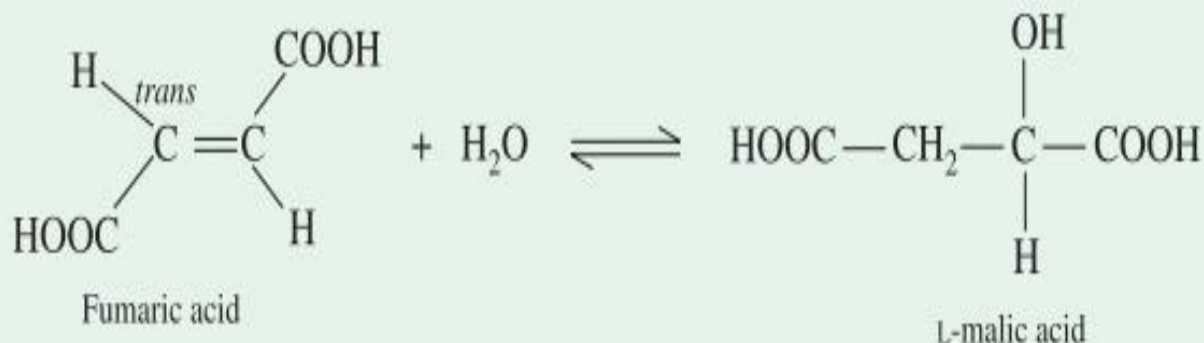
Maltase,Lactase,sucrase

Group specificity:- Some other enzymes are capable of catalyzing the reaction of a structurally related group of compounds. Example: **Pepsin** is an endopeptidase enzyme, that hydrolyzes central peptide bonds in which the amino group belongs to aromatic amino acids. Like :- phenyl alanine, tyrosine and tryptophan. **Lactic dehydrogenase (LDH)** catalyzes the interconversion of pyruvic and lactic acids and also of a number of other structurally-related compounds.

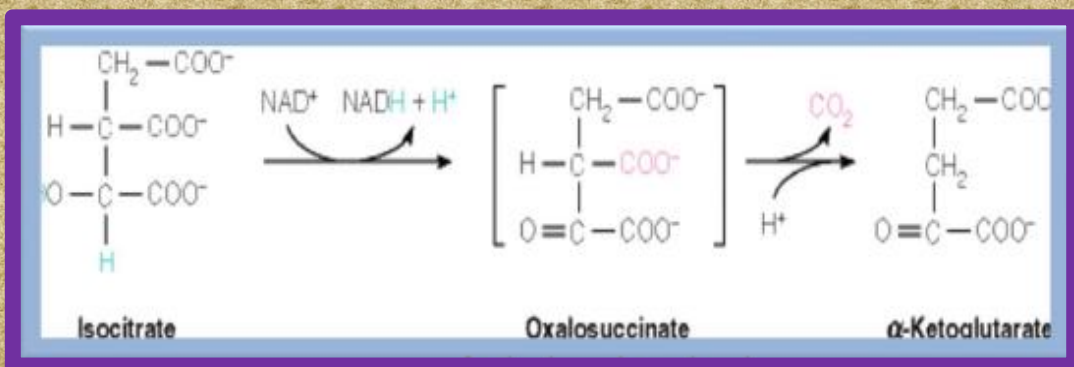
- b. **Trypsin** is an endopeptidase that hydrolyzes central peptide bonds in which the amino group belongs to basic amino acids e.g. arginine, lysine and histidine.
- c. **Chymotrypsin** is an endopeptidase that hydrolyzes central peptide bonds in which the carboxyl group belongs to aromatic amino acids.
- d. **Aminopeptidase** is an exopeptidase that hydrolyzes peripheral peptide bond at the amino terminal (end) of polypeptide chain.
- e. **Carboxypeptidase** is an exopeptidase that hydrolyzes peripheral peptide bond at the carboxyl terminal of polypeptide chain.

Optical / stereo-specificity:- The enzyme is not only specific to substrate but also to its optical configuration. Example: D-amino acid oxidase acts only on D-amino acids. L-amino acid oxidase acts only on L-amino acids. **Alpha-glycosidase, Beta-glycosidase.**

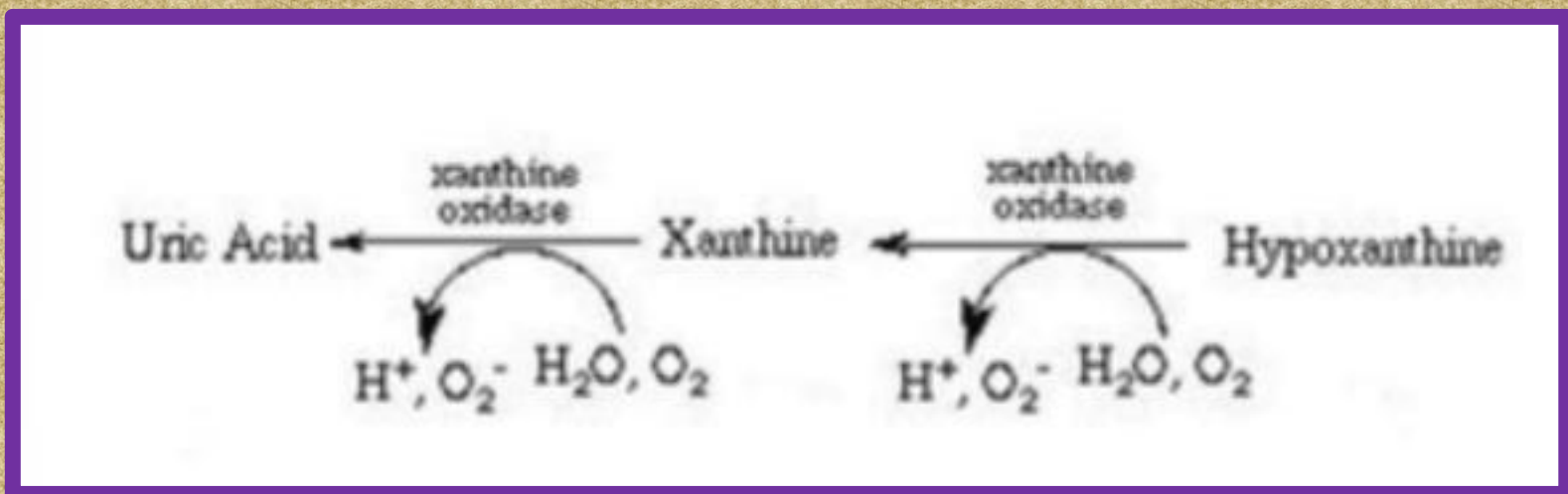
Geometrical specificity. Some enzymes exhibit specificity towards the cis and trans forms. As an example, fumarase catalyzes the interconversion of fumaric and malic acids :



DUAL SPECIFICITY:- There are two types of dual specificity. The enzyme may act on one substrate by two different reaction types. Example: **Isocitrate dehydrogenase** enzyme acts on isocitrate (one substrate) by oxidation followed by decarboxylation (two different reaction types).

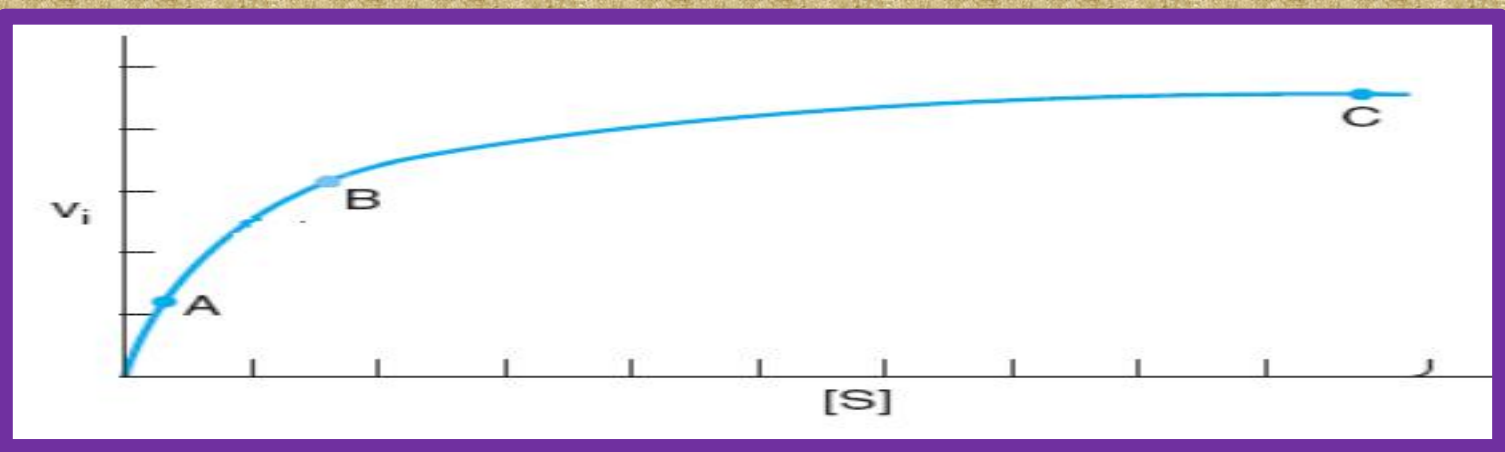


The enzyme may act on two substrates by one reaction type. Example: Xanthine oxidase enzyme acts on xanthine and hypoxanthine (two substrates) by oxidation (one reaction type).

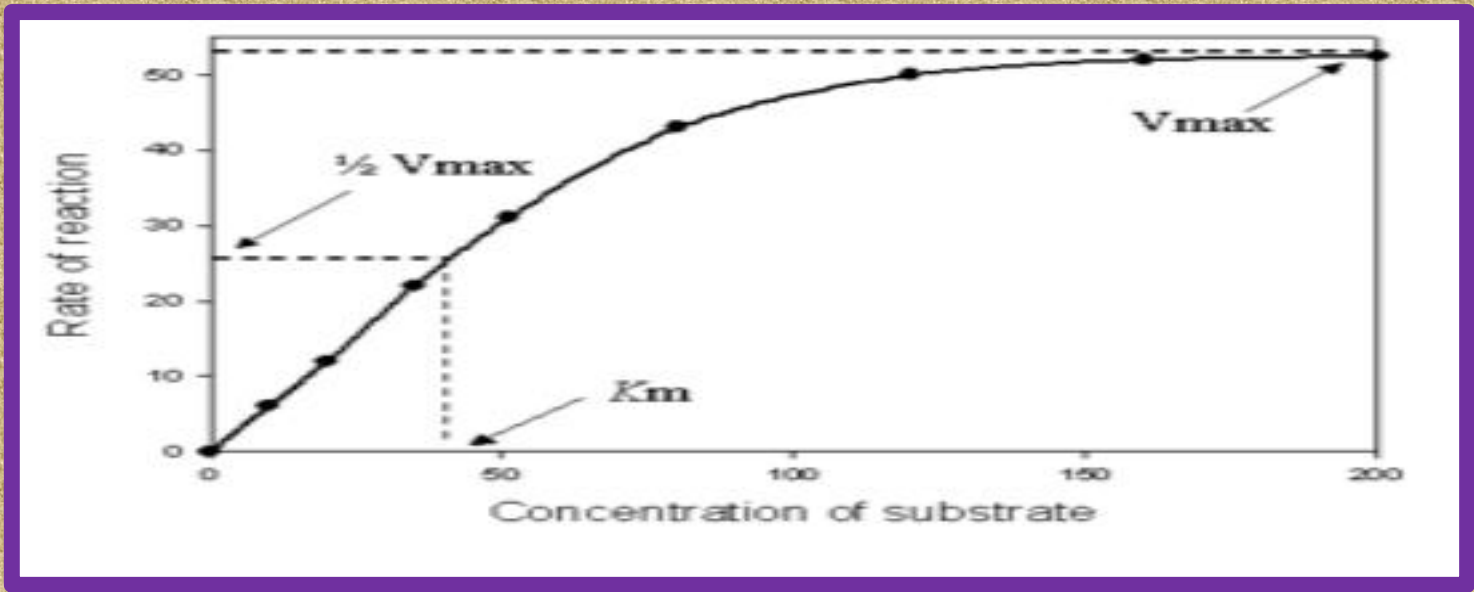


Factors affecting enzymatic reaction: The factors that mainly influence any enzyme-catalysed reaction are: Substrate concentration, Enzyme concentration, Temperature, pH, Inhibitors, Time

Substrate concentration: Increase in the substrate concentration gradually increases the velocity of enzyme reaction within the limited range of substrate levels. A rectangular hyperbola is obtained when velocity is plotted against the substrate concentration. Three distinct phases of the reaction are observed in the graph (A-linear; B-curve; C-almost unchanged). When the velocity of the reaction is almost proportional to the substrate concentration, the rate of the reaction is said to be **first order** with respect to substrate. When the substrate concentration is much greater than Concentration of enzyme, the rate of reaction is independent of substrate concentration, and the reaction is said to be **zero order**.



◆ At initial point, even though the substrate molecules are present in excess than enzyme on molar basis, not all the enzyme molecules present combine with the substrate. Hence, increasing the substrate concentration will increase the amount of enzyme associated with substrate as ES and thus v will depend on $[S]$. At V_{max} , all the enzyme molecules are saturated with substrate molecules so that further increase in $[S]$ cannot result in increased reaction rate. Michaelis-Menten derived an equation to explain this type of behaviour.

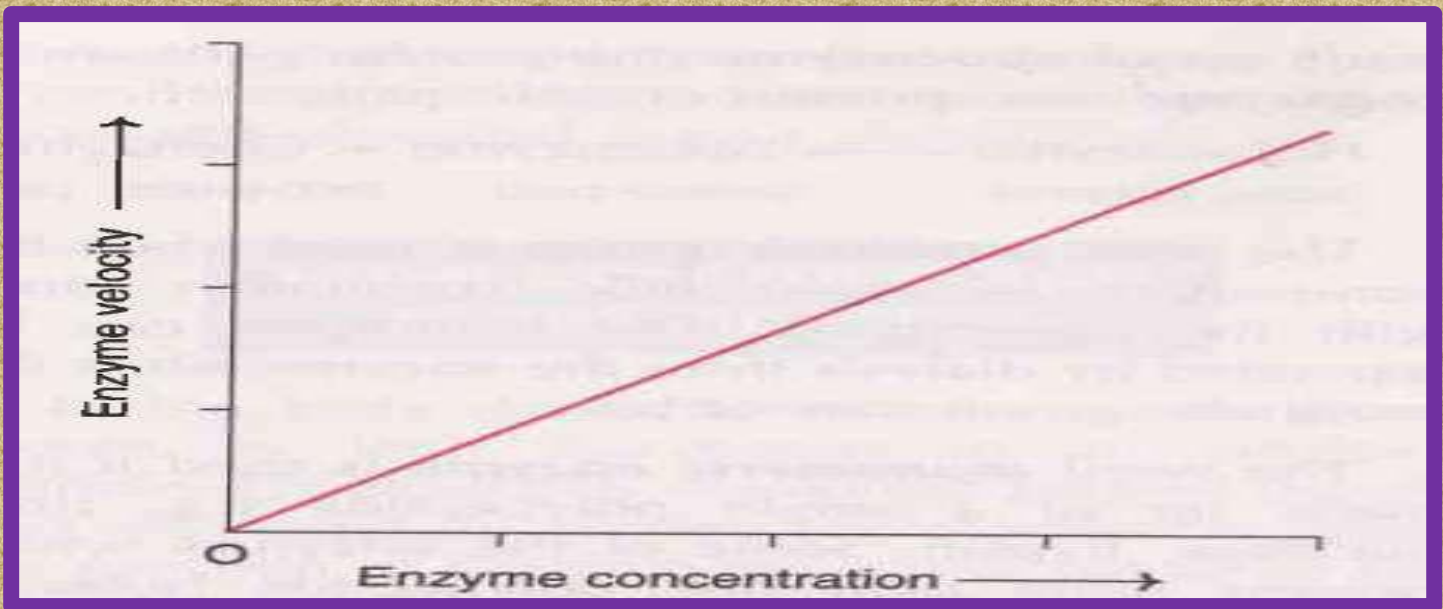


◆

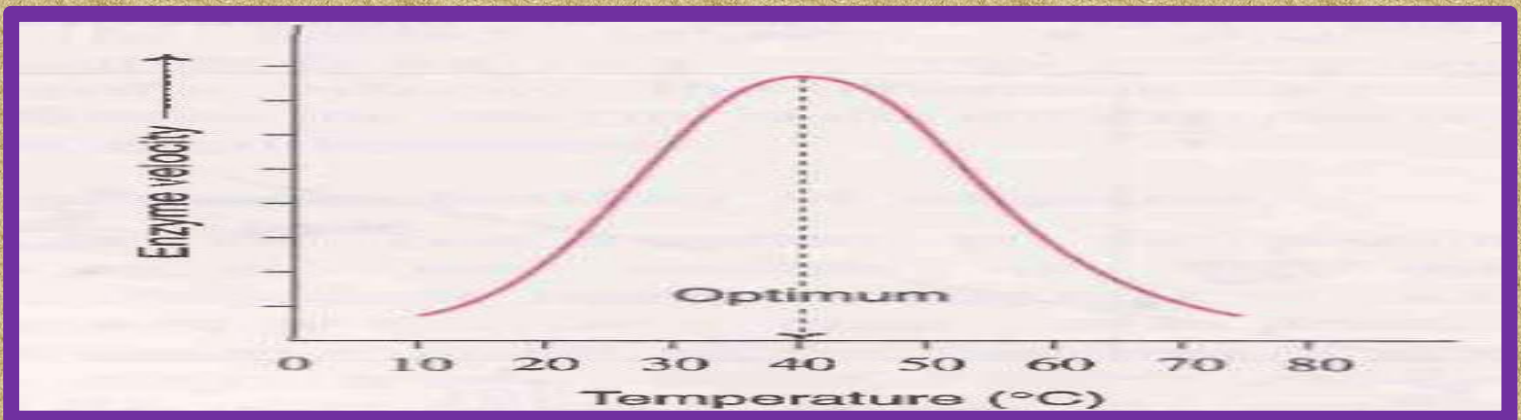
$$v = \frac{V_{max} [S]}{K_m + [S]}$$

Michaelis-Menten
equation

Enzyme concentration: When compared to substrate concentration, the concentration of enzyme is always very very low on molar basis. As the concentration of the enzyme is increased, the velocity of the reaction proportionately increases. This property of enzyme is made use in determining the serum enzymes for the diagnosis of diseases.



Temperature: Velocity of an enzyme reaction increases with increase in temperature up to a maximum and then declines. A bell-shaped curve is usually observed.



The velocity of enzyme-catalysed reactions roughly doubles with a 10°C rise in temperature over a limited range of temperature. Enzymes, being proteins, are denatured by heat and become inactive as the temperature increases beyond a certain point. Most of the enzymes are inactivated at temperatures above 60°C. The temperature at which the reaction rate is maximum is known as **optimum temperature**.

pH: Most enzymes have a characteristic pH at which their activity is maximum; above or below this pH, the activity declines. The pH affects the ionic state of the enzyme and frequently that of the substrate also. If a negatively charged enzyme (E^-) reacts with a positively charged substrate (SH^+), ESH is formed. At low pH values, E^- will be protonated and ESH is not formed. Similarly, at very high pH values SH^+ will ionize and lose its positive charge.

Effect of Time: Under ideal and optimal conditions (like pH, temperature etc.), the time required for an enzyme reaction is less. Variations in the time of the reaction are generally related to the alterations in pH and temperature.

Effect of light and radiation: Exposure of enzymes to ultraviolet, beta, gamma and X-rays inactivates certain enzymes. The inactivation is due to the formation of peroxides.

Example: UV rays inhibit salivary amylase activity.

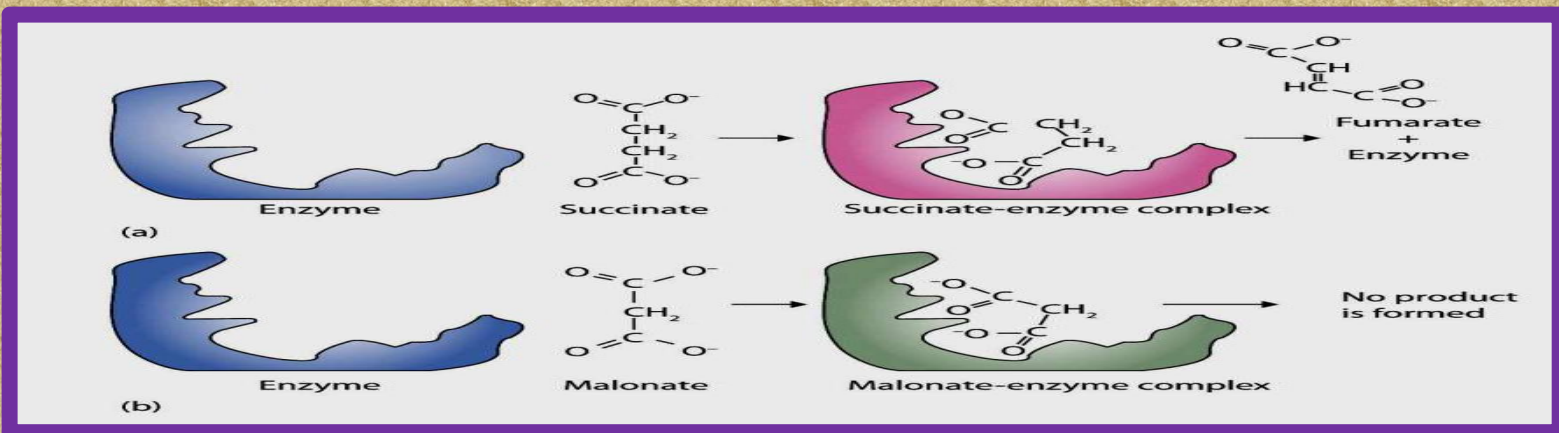
Enzyme Inhibitors: Inhibitor is a molecule that attaches itself to an enzyme and interferes with the enzyme's ability to form an enzyme-substrate complex. Specific small molecules or ions can inhibit even non-allosteric enzymes. In irreversible inhibition, the inhibitor is covalently linked to the enzyme or bound so tightly that its dissociation from the enzyme is very slow. Covalent inhibitors provide a means of mapping the enzyme's active site. In contrast, reversible inhibition is characterized by a more rapid equilibrium between enzyme and inhibitor. A chemical substance which can react in place of substrate with the enzyme but is not transferred into products and block the active site of the enzyme temporarily or permanently is called inhibitor.

There are two types of inhibitor: **Irreversible inhibitor**, **Reversible inhibitor**

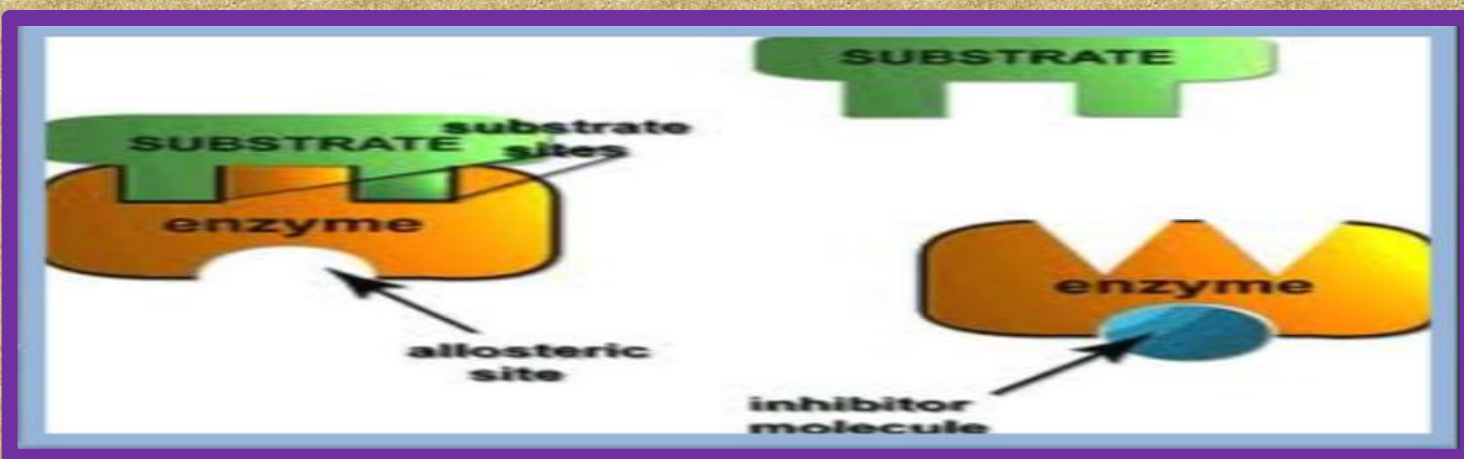
Irreversible inhibitors: This type of inhibition involves the covalent attachment (permanently) of the inhibitor to the enzyme. The catalytic activity of enzyme is completely lost. It can only be restored only by synthesizing molecules. Example: -Very small concentrations of chemical reagents such as the heavy metal ions; -Mercury, Silver, Arsenic or certain iodine-containing compounds completely inhibit some enzymes.

Reversible inhibitor: They form weak linkages with the enzymes. Their effect is completely or partly neutralized by increasing the concentration of the substrate. They are further divided into two major types: **Competitive** and **non-competitive**

Competitive: Is any compound which closely resembles the chemical structure and molecular geometry of the substrate. Competes for the same active site as the substrate molecule. Usually reversible if sufficient substrate molecules are available to ultimately displace the inhibitor. Therefore, the amount of enzyme inhibition depends upon:- Inhibitor concentration, Substrate concentration, and Relative affinities of the inhibitor and substrate for the active site.



Uncompetitive Inhibitor: In this type of inhibition, inhibitor does not compete with the substrate for the active site of enzyme instead it binds to an other site known as allosteric site.



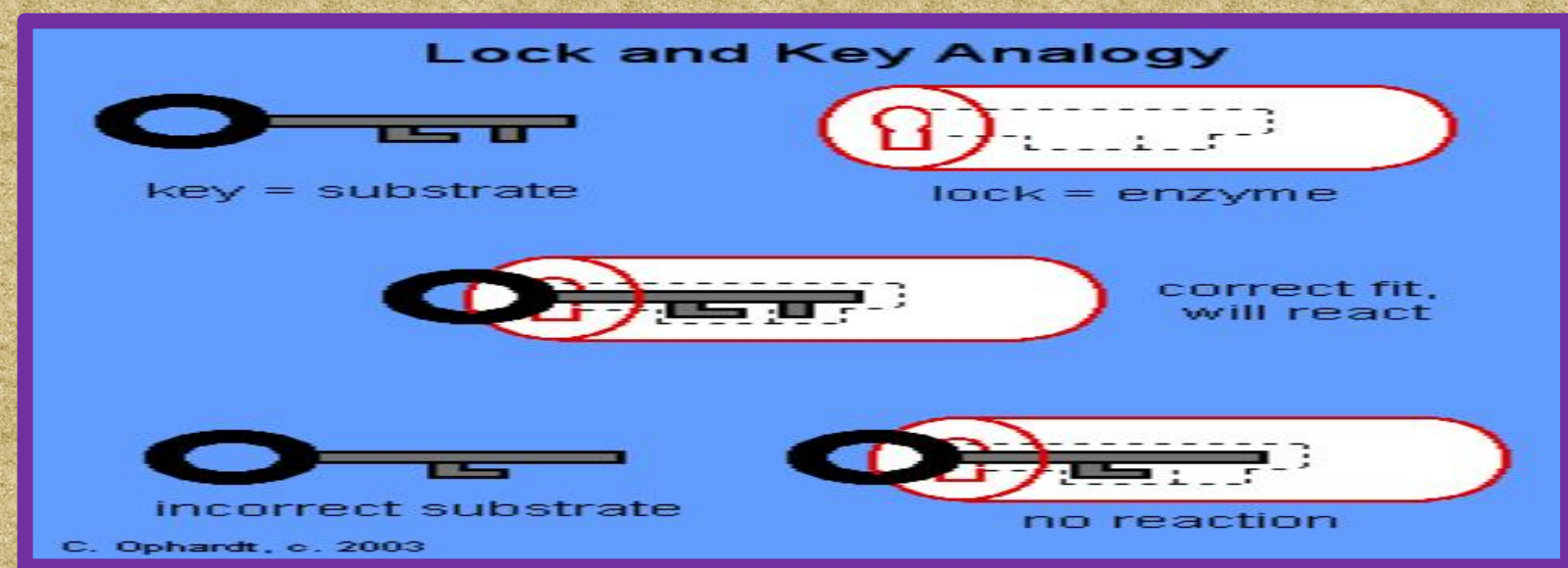
Mechanism of Enzyme Action

The active site of enzyme is the region that binds substrates, cofactors and prosthetic groups and contains residue that helps to hold the substrate. Active sites generally occupy less than 5% of the total surface area of enzyme. Active site has a specific shape due to tertiary structure of protein. A change in the shape of protein affects the shape of active site and function of the enzyme.

Active site can be further divided into: **Binding Site**: it chooses the substrate and binds it to active site **Catalytic Site**: performs the the catalytic action of enzyme .Two theories were proposed to explain the mechanism of enzyme action or how enzyme substrate complex is formed.

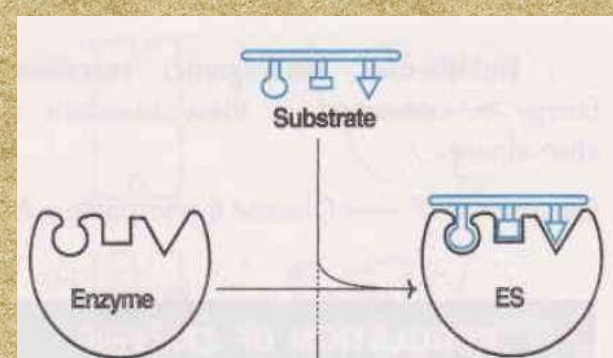
1. **Lock and key model/ Fischer's template Theory**
2. **Induced fit theory/Koshland's model**

Lock and key model/ Fischer's template Theory: Proposed by a Emil Fischer. Very first model proposed to explain an enzyme catalyzed reaction According to this model, the structure or conformation of the enzyme is rigid. The substrate fits to the binding site just as a key fits into the proper lock or a hand into the proper glove. Thus the active site of an enzyme is a rigid and pre-shaped template where only a Specific substrate can bind.



Drawback of this model :

1. Does not give any scope for the flexible nature of enzymes
2. Totally fails to explain many facts of enzymatic reactions
3. Does not explain the effect of allosteric modulator

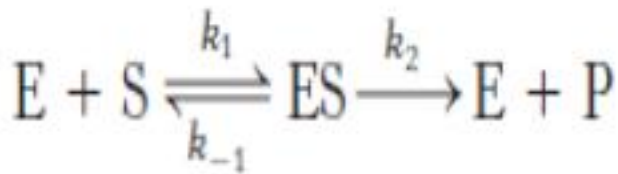


Koshland's induced-fit theory: Koshland proposed this model as the active site is not rigid and pre-shaped. The interaction of the substrate with the enzyme induces a fit or a conformation change in the enzyme, resulting in the formation of a strong substrate binding site. Furthermore, the appropriate amino acids of the enzyme are repositioned to form the active site and bring about the catalysis.

This model was accepted because: Has sufficient experimental evidence from the X-ray diffraction studies. This model also explains the action of allosteric modulators and competitive inhibition on enzymes.

Enzyme kinetics

The Michaelis-Menten Model: The kinetic properties of many enzymes are described by the Michaelis-Menten model. In this model, an enzyme (E) combines with a substrate (S) to form an enzyme-substrate (ES) complex, which can proceed to form a product (P) or to dissociate into E and S.



The rate V_0 of formation of product is given by the Michaelis-Menten equation:

$$V_0 = V_{\max} \frac{[S]}{[S] + K_M}$$

In which $V_{\max}$ is the reaction rate when the enzyme is fully saturated with substrate and K_M , the Michaelis constant, is the substrate concentration at which the reaction rate is half maximal. Or It is a measure of the affinity of an enzyme for its substrate.

The maximal rate, $V_{\max}$, is equal to the product of k_2 or k_{cat} and the total concentration of enzyme. The kinetic constant k_{cat} , called the turnover number, is the number of substrate molecules converted into product per unit time at a single catalytic site when the enzyme is fully saturated with substrate. Turnover numbers for most enzymes are between 1 and 10^4 per second. The ratio of k_{cat}/K_M provides a penetrating probe into enzyme efficiency.

Allosteric enzymes constitute an important class of enzymes whose catalytic activity can be regulated. These enzymes, which do not conform to Michaelis-Menten kinetics, have multiple active sites. These active sites display cooperativity, as evidenced by a sigmoidal dependence of reaction velocity on substrate concentration.

Thank you!!!

